

CMP-26 - Recovery Coordinator

Coordinate containment, recovery plans, identity and state reconciliation, restoration, rollback, compensation, and continuity evidence.

Authority and state ownership

Book I: Decisions 35, 38-43, 47-51, and 58. Book II: Chapters 27, 28. Authoritative state: Recovery Cases, recovery plans, reconciliation decisions, restoration progress, RPO and RTO evidence. Explicitly non-owned: domain authoritative state, constitutional admission, incident command, backup storage.

Critical invariant

Recovery restores verified identity, authority, state, evidence, and trust; restart alone is not Recovery.

Logical interfaces

CMP-26-IF-01 / Command / Open Recovery Case

Command `Open Recovery Case` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-26-IF-02 / Command / Authorize Recovery Step

Command `Authorize Recovery Step` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-26-IF-03 / Command / Restore State

Command `Restore State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-26-IF-04 / Command / Reconcile Partition

Command `Reconcile Partition` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-26-IF-05 / Command / Close Recovery

Command `Close Recovery` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-26-IF-06 / Query / Get Recovery State

Query `Get Recovery State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-26-IF-07 / Query / Get RPO/RTO Result

Query `Get RPO/RTO Result` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-26-IF-08 / Query / Explain Reconciliation

Query `Explain Reconciliation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-26-IF-09 / Event / Recovery Started

Event `Recovery Started` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-26-IF-10 / Event / State Restored

Event `State Restored` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-26-IF-11 / Event / Partition Reconciled

Event `Partition Reconciled` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-26-IF-12 / Event / Recovery Failed

Event `Recovery Failed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Detected > Contained > Planned > Restoring > Validating > Closed or Escalated. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: wrong recovery point, identity discontinuity, partition conflict, evidence loss, premature service return. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-26-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-26-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-26-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-26-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-26-TST-005 - Critical invariant. Recovery restores verified identity, authority, state, evidence, and trust; restart alone is not Recovery. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-26-TST-006 - Failure containment. Inject wrong recovery point, identity discontinuity, partition conflict, evidence loss, premature service return. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-26-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-26-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-26-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-26-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.